

Creating Public and Private Subnets

To Begin with the Lab

Summary of the Lab

This lab demonstrated how to create a highly available network architecture by configuring two public and two private subnets within an existing custom VPC across different Availability Zones. An Internet Gateway was created and attached to the VPC to enable internet connectivity. A dedicated public route table was then created, associated only with the public subnets, and configured with a default route (0.0.0.0/0) pointing to the Internet Gateway. The private subnets remained associated with the main route table without direct internet access, ensuring secure network isolation while following AWS best practices for scalable and resilient infrastructure.

- Note: This lab assumes that the custom VPC (192.168.0.0/24) has already been created. Begin directly with the subnet creation process.
- Please delete the previously created subnets and Internet Gateway as well. We'll recreate them here.
- Open the **AWS Management Console** and navigate to the **VPC** service. Since the VPC is already available, proceed to the **Subnets** section to create separate public and private subnets following AWS best practices.
- Click Create Subnets and select your existing VPC. Instead of creating one subnet at a time, create all four subnets in a single operation so that the VPC is prepared for a highly available application deployment.
- Create the first subnet by naming it Public-Subnet-1, selecting the us-east-1a Availability Zone, and assigning the CIDR block 192.168.0.0/26. This subnet will later be configured as a public subnet.

Subnet 1 of 4

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public-Subnet-1

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / us-east-1a (us-east-1a)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.0/26

64 IPs

- Add a second subnet named Public-Subnet-2, choose the us-east-1b Availability Zone, and assign the CIDR block 192.168.0.64/26. Placing the subnet in a different Availability Zone improves application availability.

Subnet 2 of 4

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public-Subnet-2

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az6 (us-east-1b)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.64/26

64 IPs

- Add a third subnet named Private-Subnet-1, select the us-east-1a Availability Zone, and assign the CIDR block 192.168.0.128/26. This subnet will later host private resources.

Subnet 3 of 4

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Private-Subnet-1

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az4 (us-east-1a)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.128/26

64 IPs

- Create the fourth subnet by naming it Private-Subnet-2, selecting the us-east-1b Availability Zone, and assigning the CIDR block 192.168.0.192/26. After verifying all subnet details

Subnet 4 of 4

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Private-Subnet-2

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az6 (us-east-1b)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.192/26

64 IPs

- Click Create Subnets to create all four subnets.

Creating subnets...

We are currently creating subnets.

Creating subnet 4

75%

▼ Details

Creating subnet 1

Creating subnet 2

Creating subnet 3

Creating subnet 4

- Four Subnets created Successfully

☐	Name	Subn...	State	VPC	Bloc...	IPv4 CIDR	IPv6 ...	IPv6 ...	Avail...	Availability Zone
☐	Public-Subnet-2	subnet-...	Available.	vpc-02a...	Off	192.168.0.64/26	-	-	59	use1-az6 (us-east-1b)
☐	Private-Subnet-1	subnet-...	Available.	vpc-02a...	Off	192.168.0.128/26	-	-	59	use1-az4 (us-east-1a)
☐	Public-Subnet-1	subnet-...	Available.	vpc-02a...	Off	192.168.0.0/26	-	-	59	use1-az4 (us-east-1a)
☐	Private-Subnet-2	subnet-...	Available.	vpc-02a...	Off	192.168.0.192/26	-	-	59	use1-az6 (us-east-1b)

- Navigate to the **Internet Gateways** section and create a new Internet Gateway by providing a suitable name such as **IGW**. After the gateway is created, attach it to the existing VPC so that the VPC can communicate with the internet.

igw-0590afd31ae80d467 / IGW

Actions

Details

Internet gateway ID

igw-0590afd31ae80d467

State

Attached

VPC ID

vpc-02a0440be172ac2c9 | myfirst-vpc

Owner

463646775279

Tags (1)

Manage tags

Search tags

Key	Value
Name	IGW

- Open the **Route Tables** section and identify the **Main Route Table** that AWS automatically creates for every VPC. Verify that all four newly created subnets are automatically associated with this route table.

Route tables (1/1)

Last updated
3 minutes ago

Actions

Create route table

Find route tables by attribute or tag

VPC = vpc-02a0440be172ac2c9

Clear filters

☒	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒	-	rtb-039041c66222da988	-	-	Yes	vpc-02a0440be172ac2c9

rtb-039041c66222da988

Subnets without explicit associations (4)

Edit subnet associations

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Public-Subnet-2	subnet-0a8d7a83e629f555	192.168.0.64/26	-
Private-Subnet-1	subnet-0d66e2276cf7cfe6	192.168.0.128/26	-
Public-Subnet-1	subnet-07dba1f436052d9b5	192.168.0.0/26	-
Private-Subnet-2	subnet-0956e5acafeabb8a2	192.168.0.192/26	-

- Create a new Route Table by clicking **Create Route Table**, assigning a name such as **RT-Public**, selecting the existing VPC, and creating the route table. This route table will be used only for the public subnets.

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

- Open the **Subnet Associations** tab of the newly created **RT-Public** route table and click **Edit Subnet Associations**. Select **Public-Subnet-1** and **Public-Subnet-2**, then save the association so that only the public subnets use this route table.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Public-Subnet-2	subnet-0a8d7a83e629f555	192.168.0.64/26	-	Main (rtb-039041c66222da988)
☐	Private-Subnet-1	subnet-0d6e2276cf7cdf6	192.168.0.128/26	-	Main (rtb-039041c66222da988)
☒	Public-Subnet-1	subnet-07dba1f436052d9b5	192.168.0.0/26	-	Main (rtb-039041c66222da988)
☐	Private-Subnet-2	subnet-0956e5acafeabb8a2	192.168.0.192/26	-	Main (rtb-039041c66222da988)

Selected subnets

subnet-07dba1f436052d9b5 / Public-Subnet-1 subnet-0a8d7a83e629f555 / Public-Subnet-2

- Open the **Routes** tab of **RT-Public**, click **Edit Routes**, and add a new route with the destination **0.0.0.0/0**. Select the attached **Internet Gateway** as the target and save the changes to provide internet access for the public subnets.

Edit routes

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/24	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Verify that the route is in the **Active** state and that the Internet Gateway is attached to the VPC. The public subnets now have internet connectivity, while the private subnets remain associated with the Main Route Table and continue to have no direct internet access.

Routes					
Subnet associations					
Edge associations					
Route propagation					
Tags					
Routes (2)					
Q Filter routes					
Both Edit routes					
< 1 > ⚙					
Destination	Target	Status	Propagated	Route Origin	
0.0.0.0/0	igw-0590afd31ae80d467	Active	No	Create Route	
192.168.0.0/24	local	Active	No	Create Route Table	